Assignment 2

Network Scanning & Threat Analysis

Submitted by: Stuti Verma

Submitted to: Meghna Maan

Date : 02/03/2026

1.Introduction

This assignment focuses on understanding network reconnaissance, vulnerability and threat analysis using Nmap and VirusTotal tools. The objective is to identify open ports, running services, product or version information, and host or system details through ethical scanning techniques and analyse potential security risks.

Nmap (Network Mapper) is an open-source network scanning tool used to discover devices, open ports, and services running on a system. It helps security professionals understand network exposure and identify possible security risks through ethical scanning.

VirusTotal is an online security intelligence platform used to analyse domains, IP addresses, URLs and files using multiple antivirus engines. It provides passive information about a target without directly interacting with it.

In this assignment, Nmap is used for active network scanning, while VirusTotal is used for gathering passive threat intelligence to analyse system security.

2.Nmap Commands Used

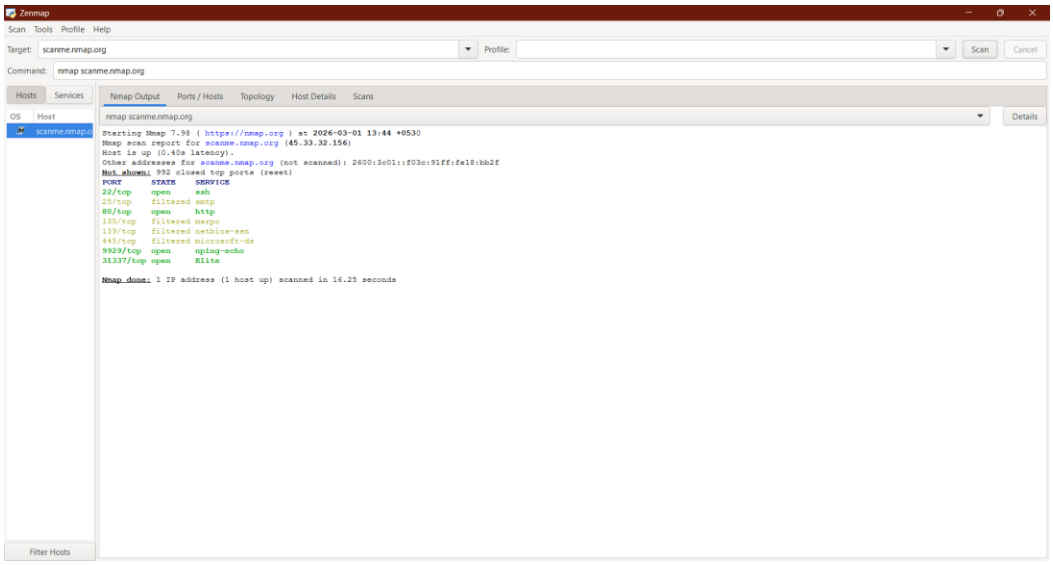
Command 1: Basic Network Scan

nmap scanme.nmap.org

Explanation:

This command performs a basic network scan to identify open ports and active services running on the target system.

Screenshot:



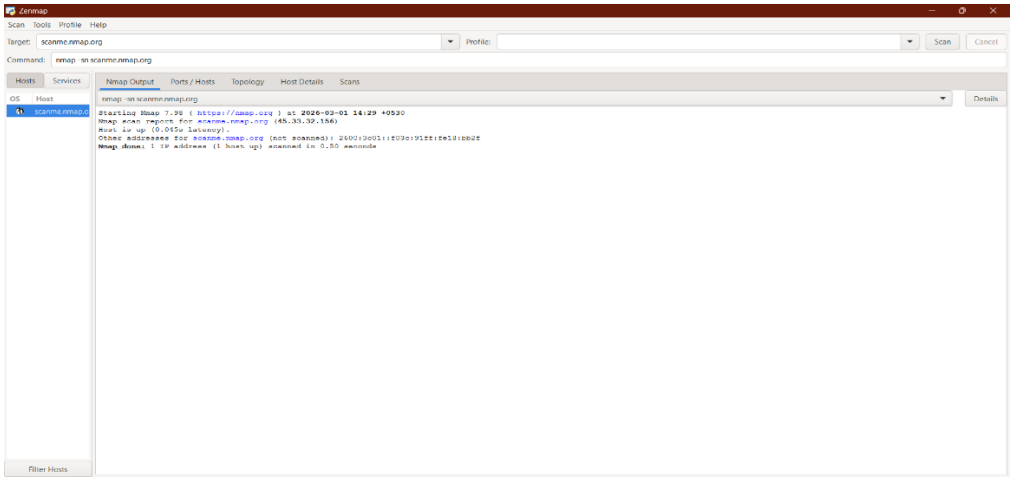
Command 2: Ping Scan (No port Scan)

nmap -sn scanme.nmap.org

Explanation:

This command checks whether the host is alive or reachable on the network without scanning ports.

Screenshot:



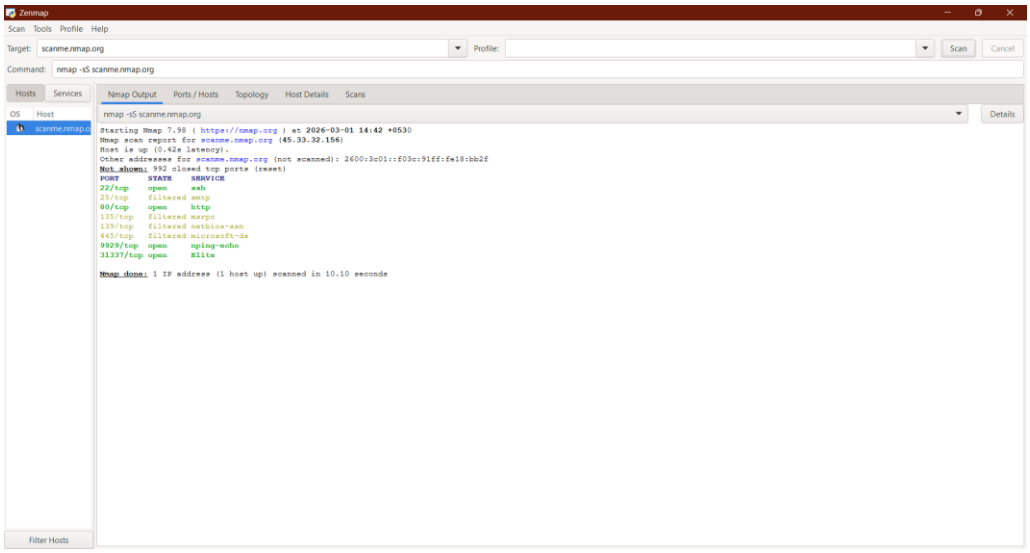
Command 3: Stealth Scan

nmap -sS scanme.nmap.org

Explanation:

This command performs a SYN scan to find open ports on host without completing the full TCP connection.

Screenshot:



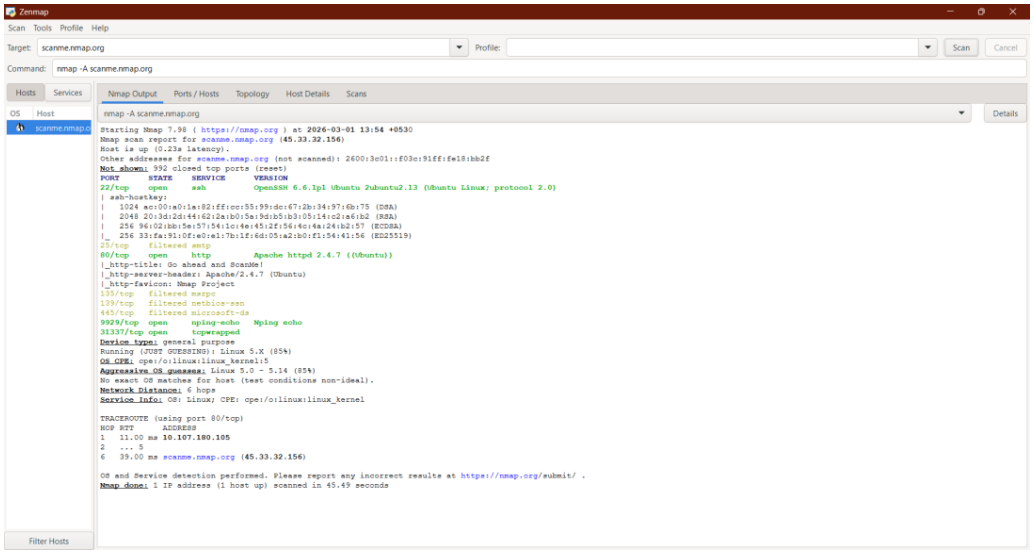
Command 4: Fast Scan

nmap -F scanme.nmap.org

Explanation:

This command performs a fast scan by checking only the most common or basic ports instead of checking all the ports.

Screenshot:



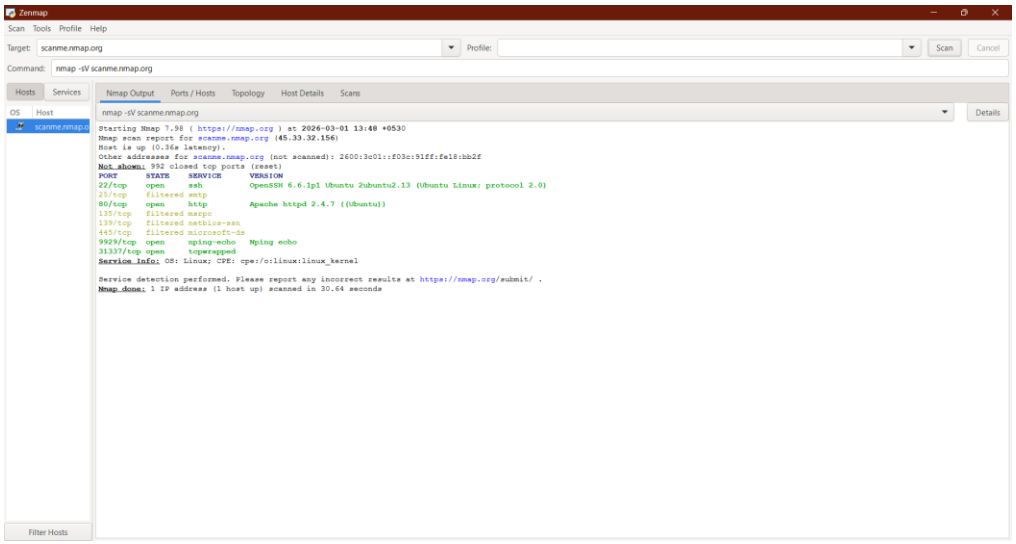
Command 5: Service Version Detection

nmap -sV scanme.nmap.org

Explanation:

This command scans the target and detects open ports along with the version of services running on those ports.

Screenshot:



Command 6: Port Range Specification

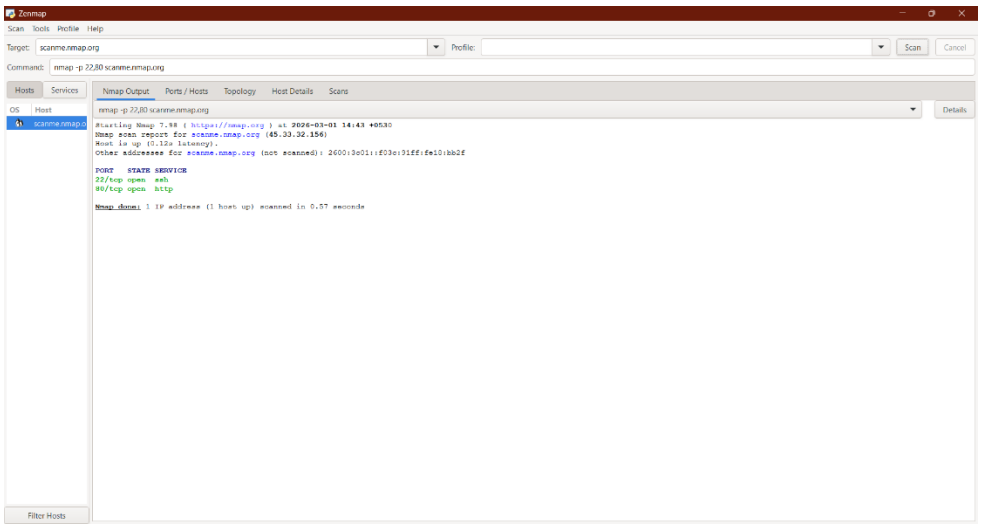
nmap -p 22,80 scanme.nmap.org

Explanation:

This command scans only specific ports, here:

- Port 22: SSH (Secure Shell)
- Port 80: HTTP (Web Services)

Screenshot:



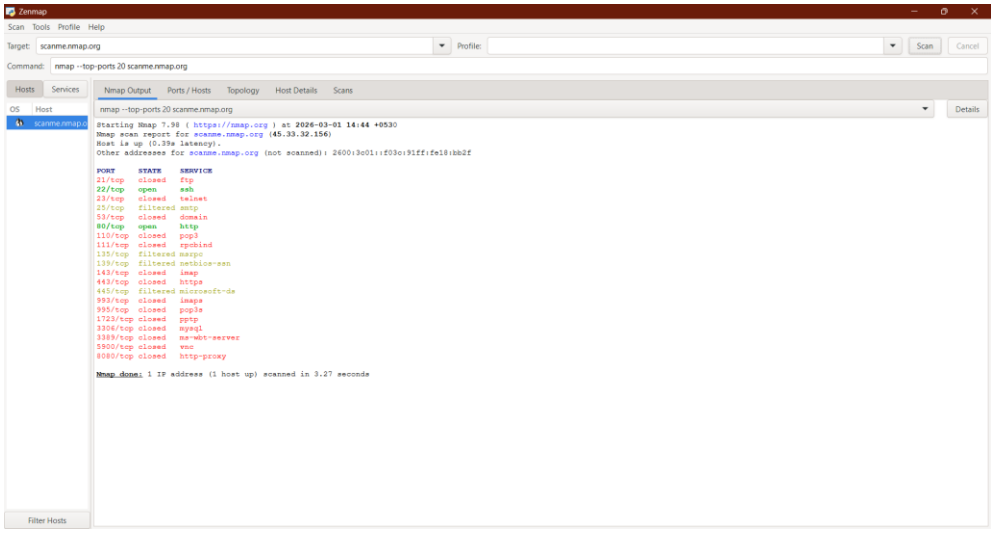
Command 7: Scan most frequently used ports

nmap --top-ports20 scanme.nmap.org

Explanation:

This command scans the top 20 most commonly used ports on host.

Screenshot:



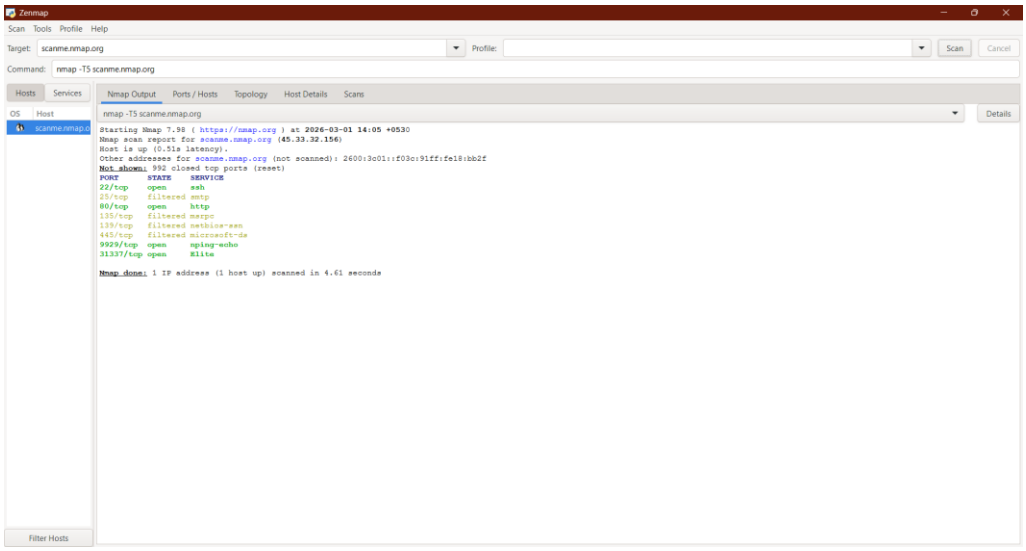
Command 8: Time Templates (T0 – T5)

nmap -T5 scanme.nmap.org

Explanation:

The command -T5(Time template level 5) performs the scan at very high speed by increasing the scanning rate.

Screenshot:



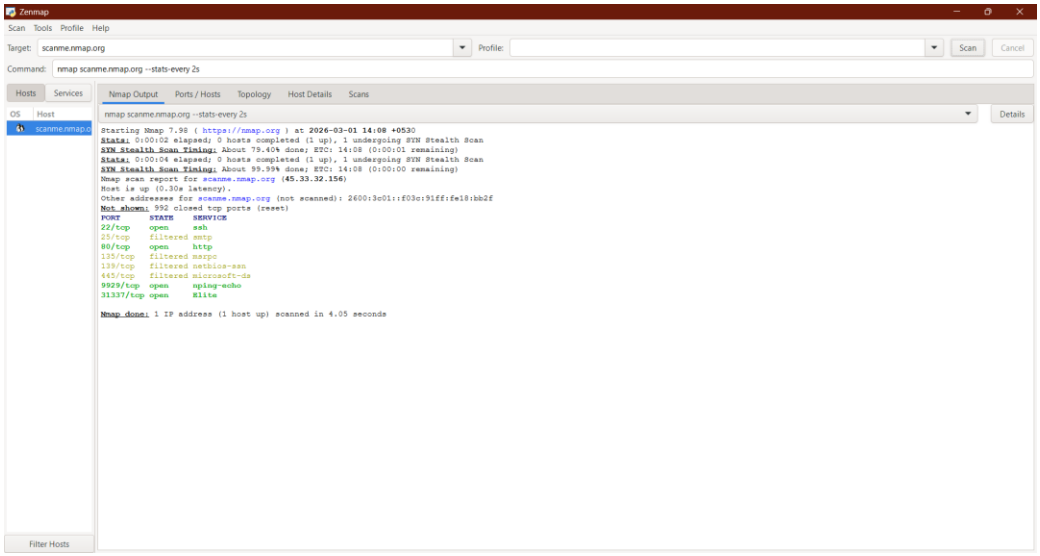
Command 9: Shows Scan Progress Statistics

nmap scanme.nmap.org --stats-every 2s

Explanation:

This command scans host and displays scan progress updates every 2 seconds.

Screenshot:



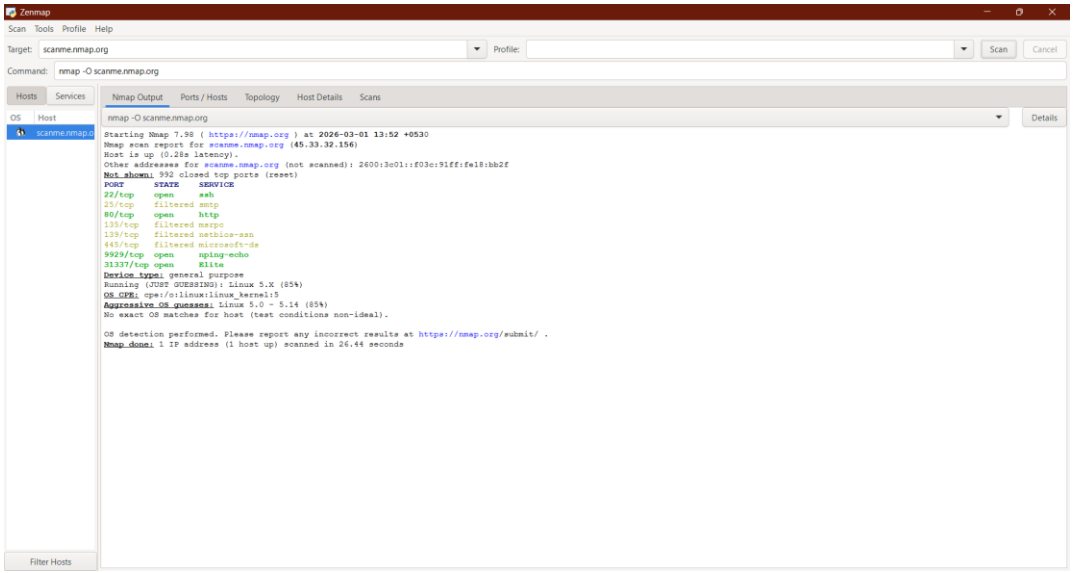
Command 10: Operating System Detection

nmap -O scanme.nmap.org

Explanation:

This command scans the target and tries to identify the operating system (OS) running on host.

Screenshot:



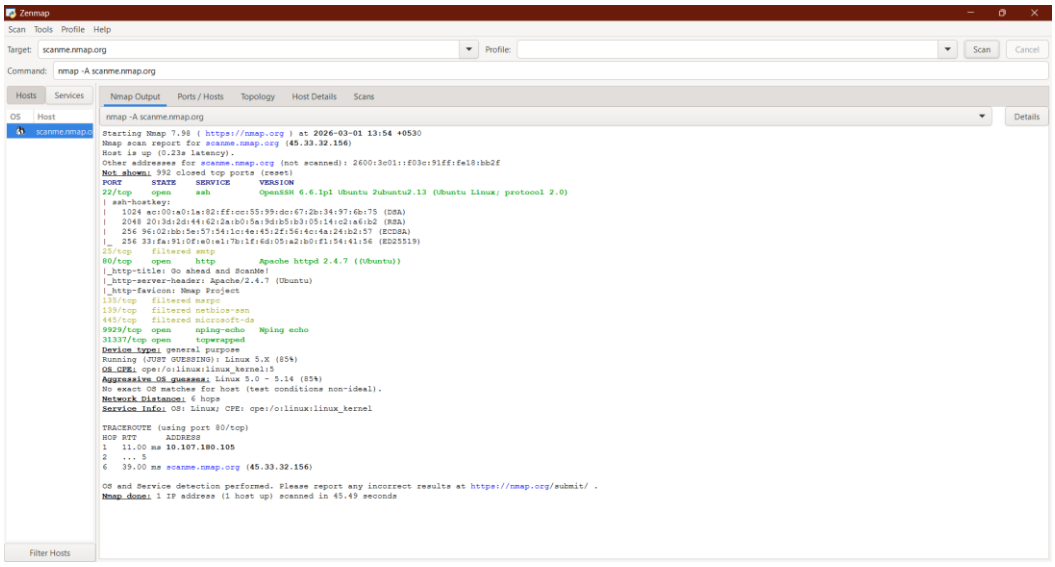
Command 11: Aggressive Scan

nmap -A scanme.nmap.org

Explanation:

This command performs an advanced and detailed scan that includes OS detection, Service Version Detection, etc. on host.

Screenshot:



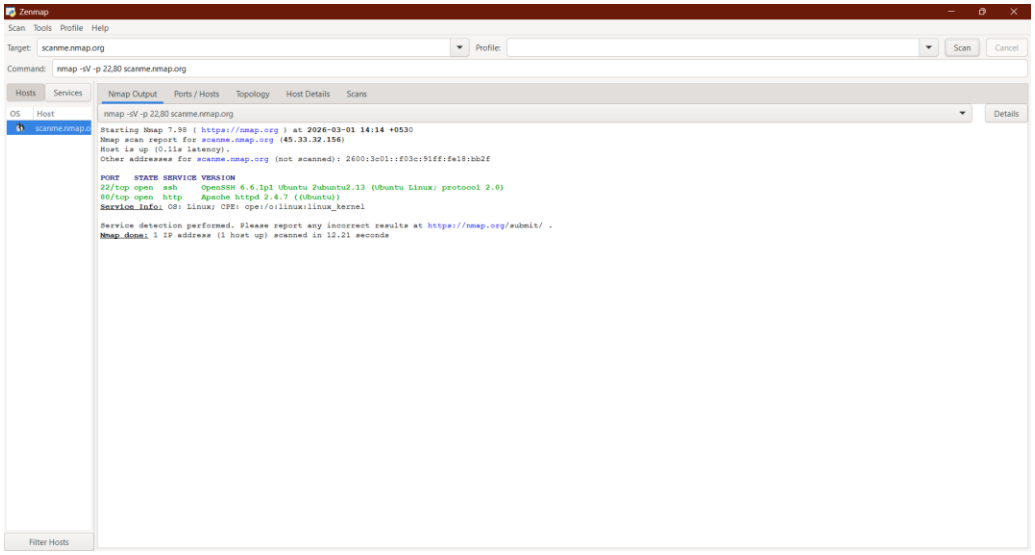
Command 12:

nmap -sV-p 22,80 scanme.nmap.org

Explanation:

This command scan only ports 22 and 80 and also detects the service versions running on those ports.

Screenshot:



3. Scan Results

PORT	STATE	SERVICES	VERSION
22	Open	SSH	OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
25	Filtered	SMTP	
80	Open	HTTP	Apache httpd 2.4.7 ((Ubuntu))
135	Filtered	MSRPC	
139	Filtered	NetBIOS-SSN	
445	Filtered	Microsoft-DS	
9929	Open	Nping-Echo	Nping Echo
31337	Open	Topwrapped	

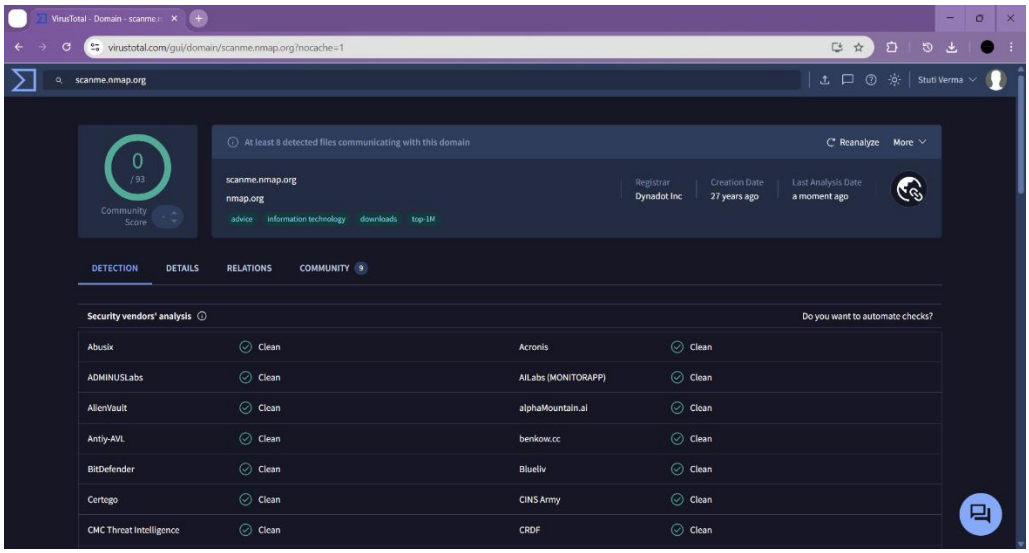
4. VirusTotal Findings

- **What is VirusTotal?**
  VirusTotal is an online security tool used to check whether a website , domain, file, or IP address is safe or malicious. It scans the given target using multiple antivirus engines and security databases to detect possible threats.

- **Uses of VirusTotal**
  In this assignment, VirusTotal is used to analyse the domain scanme.nmap.org and collect security-related information without directly scanning the system. This method is called passive intelligence gathering.

- **Findings from VirusTotal**
  - The domain reputation was found to be safe.
  - No malware or malicious activity was detected.
  - Information related to IP address and domain details was available.
  - Security vendors did not report any threats.

Detection Section:



VirusTotal - Domain - scanme.n

virustotal.com/gui/domain/scanme.nmap.org?nocache=1

scanme.nmap.org

Stuti Verma

Criminal IP	Clean	Cyble	Clean
CyRadar	Clean	desenmascara.me	Clean
DNS8	Clean	Dr.Web	Clean
EmergingThreats	Clean	Emsisoft	Clean
ESET	Clean	ESTsecurity	Clean
Forcepoint ThreatSeeker	Clean	Fortinet	Clean
G-Data	Clean	Google Safebrowsing	Clean
GreenSnow	Clean	Helmdal Security	Clean
IPsum	Clean	Juniper Networks	Clean
Kaspersky	Clean	Lionic	Clean
Malwared	Clean	MalwarePatrol	Clean
malwares.com URL checker	Clean	OpenPhish	Clean
Phishing Database	Clean	Phishtank	Clean
PREBYTES	Clean	Quick Heal	Clean
Quttera	Clean	Scantitan	Clean

VirusTotal - Domain - scanme.n

virustotal.com/gui/domain/scanme.nmap.org?nocache=1

scanme.nmap.org

Stuti Verma

SCUMWARE.org	Clean	Seclookup	Clean
securolytics	Clean	Sophos	Clean
Spam404	Clean	StopForumSpam	Clean
Sucuri SiteCheck	Clean	ThreatHive	Clean
Trustwave	Clean	URLhaus	Clean
Viettel Threat Intelligence	Clean	ViriBack	Clean
VX Vault	Clean	Webroot	Clean
Yandex Safebrowsing	Clean	ZeroCERT	Clean
0xSI_f33d	Unrated	AlphaSOC	Unrated
ArcSight Threat Intelligence	Unrated	AutoShun	Unrated
Axur	Unrated	Bfore.AI PreCrime	Unrated
Bkav	Unrated	ChainPatrol	Unrated
Chong Lua Dao	Unrated	Cluster25	Unrated
CSIS Security Group	Unrated	Cyan	Unrated
Ermes	Unrated	GCP Abuse Intelligence	Unrated

VirusTotal - Domain - scanme.n

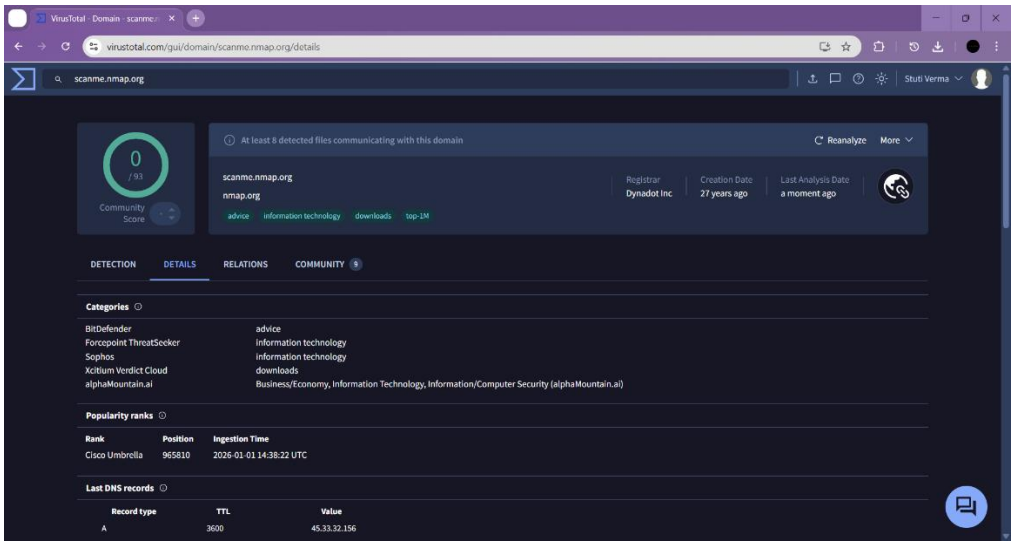
virustotal.com/gui/domain/scanme.nmap.org?nocache=1

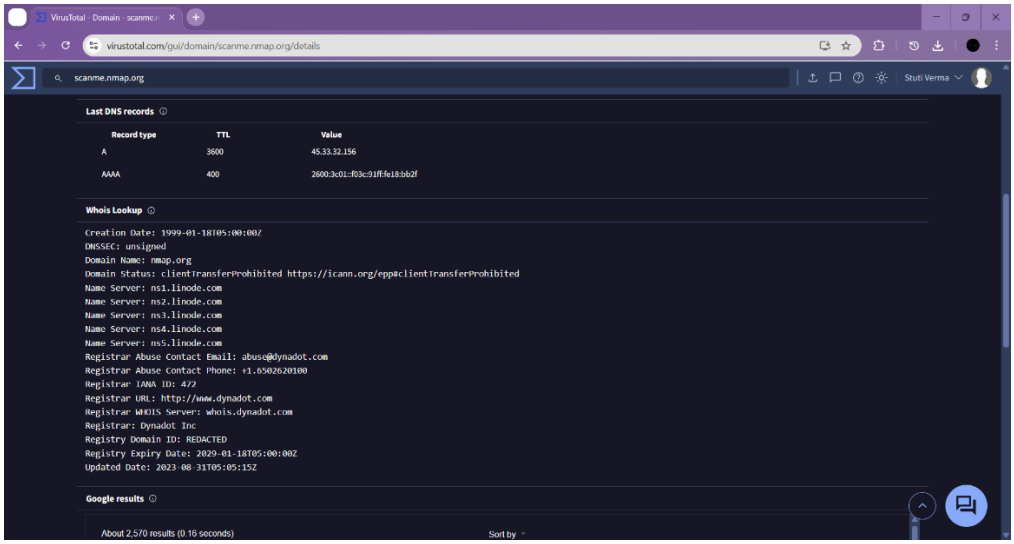
scanme.nmap.org

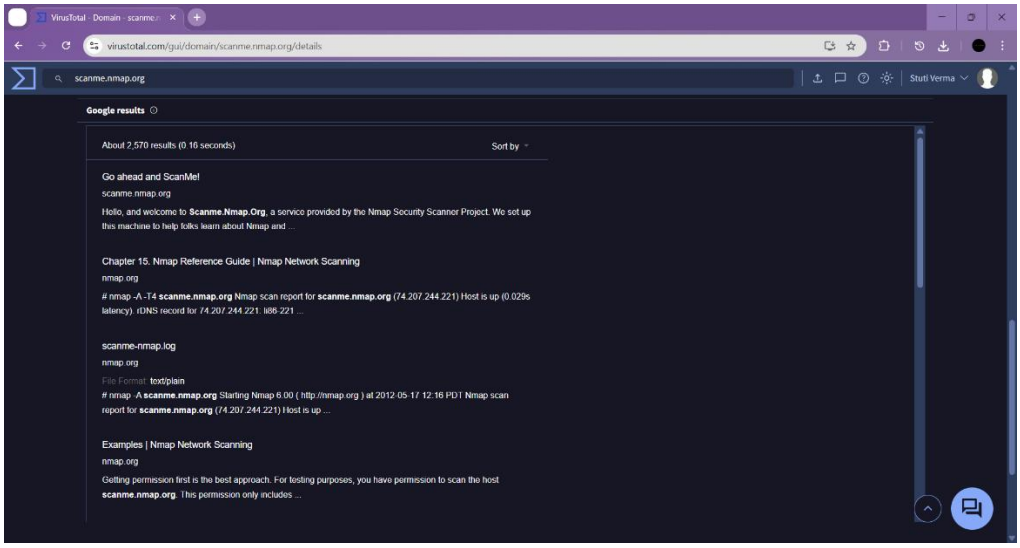
Stuti Verma

Chong Lua Dao	Unrated	Cluster25	Unrated
CSIS Security Group	Unrated	Cyan	Unrated
Ermes	Unrated	GCP Abuse Intelligence	Unrated
GreyNoise	Unrated	GridInsoft	Unrated
Hunt.io Intelligence	Unrated	Lumu	Unrated
MalwareURL	Unrated	Mimecast	Unrated
Netcraft	Unrated	PhishFort	Unrated
PhishLabs	Unrated	PrecisionSec	Unrated
SafeToOpen	Unrated	Sansec eComscan	Unrated
SecureBrain	Unrated	Snort IP sample list	Unrated
SOCRadar	Unrated	URLQuery	Unrated
VIPRE	Unrated	Xcitium Verdict Cloud	Unrated
ZeroFox	Unrated		

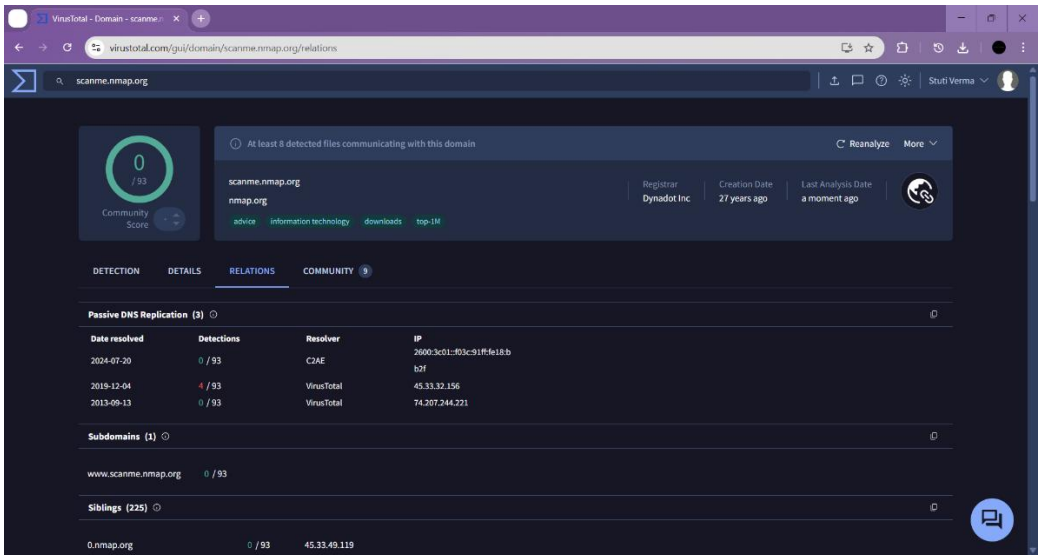
Details Section:

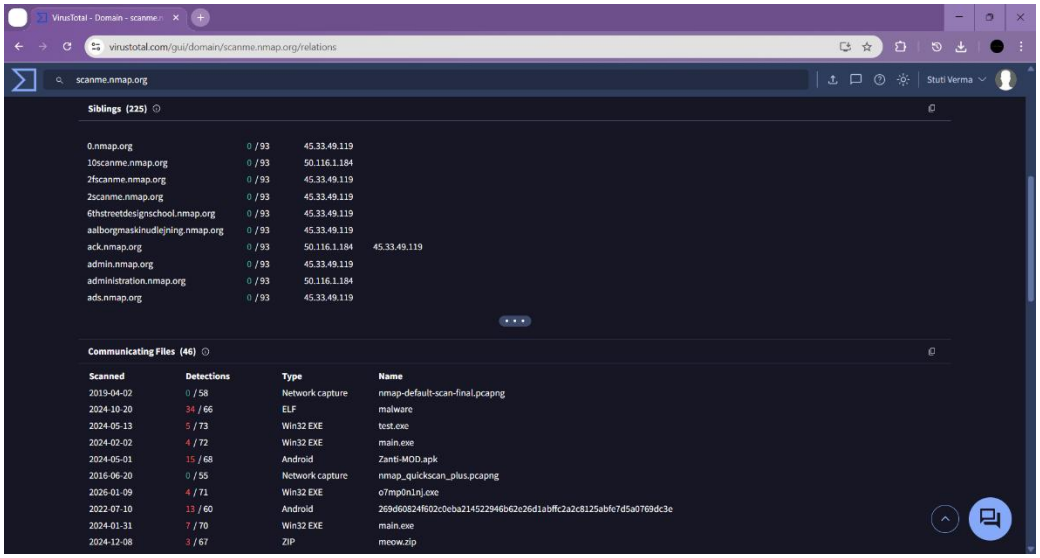


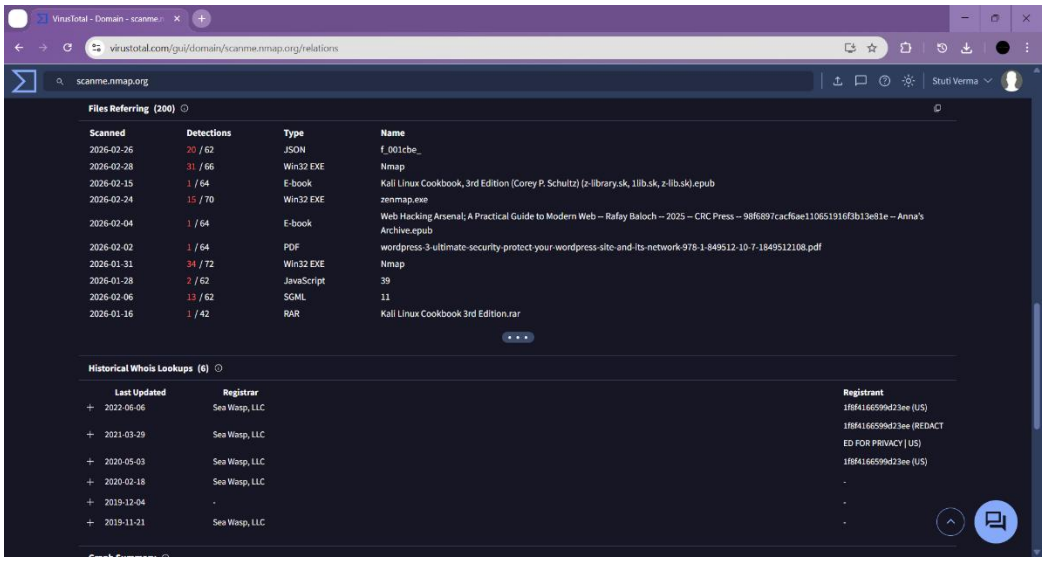


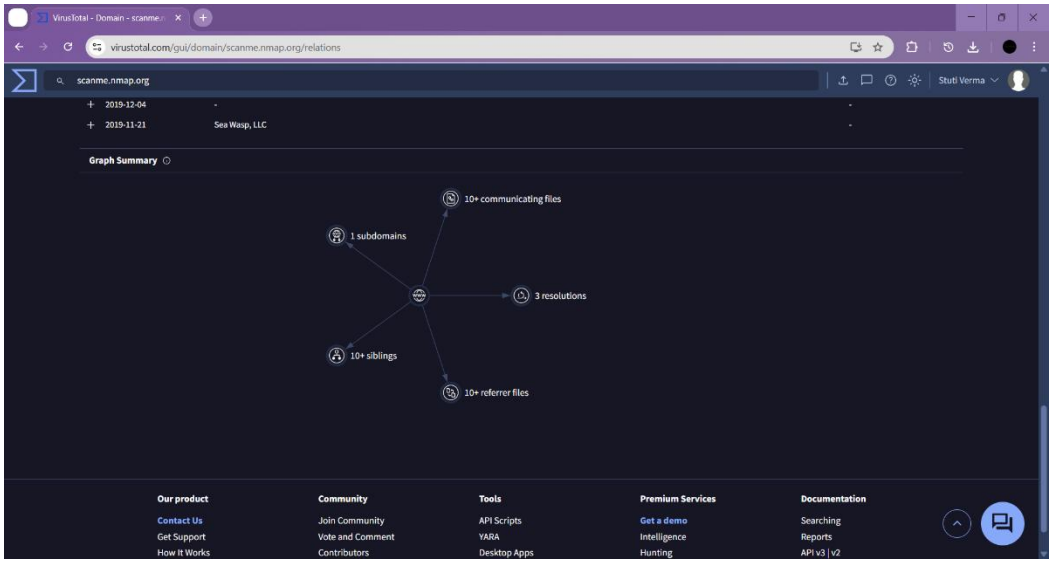


Relations Sections:








VirusTotal - Domain - scanme.n
virustotal.com/gui/domain/scanme.nmap.org/relations
scanme.nmap.org
Stuti Verma
2019-12-04
2019-11-21
Sea Wasp, LLC
Graph Summary
10+ communicating files
1 subdomains
3 resolutions
10+ siblings
10+ referrer files
Our product
Contact Us
Get Support
How It Works
Community
Join Community
Vote and Comment
Contributors
Tools
API Scripts
YARA
Desktop Apps
Premium Services
Get a demo
Intelligence
Hunting
Documentation
Searching
Reports
API v3 | v2

5. Port & Services Analysis

I. **Port 22 - SSH (Secure Shell)**

Meaning: SSH is used for secure remote login to a system. Administrators use it to access and manage servers from another computer.

Security Implication: If weak passwords are used, attackers may gain unauthorized access to the system through brute-force attacks.

II. **Port 25 – SMTP (Simple Mail Transfer Protocol)**

Meaning: SMTP is used to send emails between mail servers.

Security Implication: If not properly configured, attackers may misuse it to send spam emails or perform email spoofing attacks.

III. **Port 80 – HTTP (Hyper Text Transfer Protocol)**

Meaning: HTTP is used to access websites through web browsers.

Security Implication: Data transmitted through HTTP is not encrypted, which may allow attackers to intercept sensitive information.

IV. **Port 135 – MSRPC (Microsoft Remote Procedure Call)**

Meaning: MSRPC allows communication between different programs or systems in Windows networks.

Security Implication: This service may expose system information and can be targeted by attackers to exploit Windows vulnerabitities.

V. **Port 139 - NetBIOS-SSN**

Meaning: NetBIOS Session Service is used for file and printer sharing within local networks.

Security Implication: Improper configuration may allow attackers to access shared files or gather system information.

VI. **Port 445 – Microsoft-DS**

Meaning: Microsoft-DS supports file sharing and network communication in windows systems.

Security Implication: This port is commonly targeted by malware and ransomware attacks if security patches are not updated.

6. Vulnerability Reasoning

The Nmap scan identified several open ports and running services along with their software versions. These services are necessary for system communication, but they may also introduce security risks if not properly secured or updated.

Services such as SSH, SMTP, and HTTP are commonly used for remote access, email transfer, and web communication. If older versions of these services are running, attackers may take advantage of known vulnerabilities available publicly.

And services like MSRPC, NetBIOS, and Microsoft-DS allow system communication and file sharing within networks. However, if access control is weak or configurations are incorrect, sensitive system information or shared resources may become exposed.

The presence of open ports means the system is reachable over the network. Attackers often perform scanning to identify such services and analyse possible weaknesses. Even services used for testing, such as Nping-Echo, may reveal network response information.

Therefore, regular software updates, strong authentication methods, firewall protection, and proper service configuration are important to reduce potential security risks and maintain system safety.

7. Conclusion

This assignment helped me understand how network scanning and threat analysis are performed using tools like Nmap and VirusTotal. I learned how to identify open ports, running services, and system information through ethical scanning. This assignment also showed how exposed services can create security risks if not properly managed. Overall, it improved my knowledge and understanding of basic network security and how security analysts analyse systems to identify potential threats.